Multihoming in Symbian OS v7.0s

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- Implementation Concepts
- Application API's

Symbian OS networking architecture: A recap

Background | Implementation concepts | Application APIs

Networking architecture

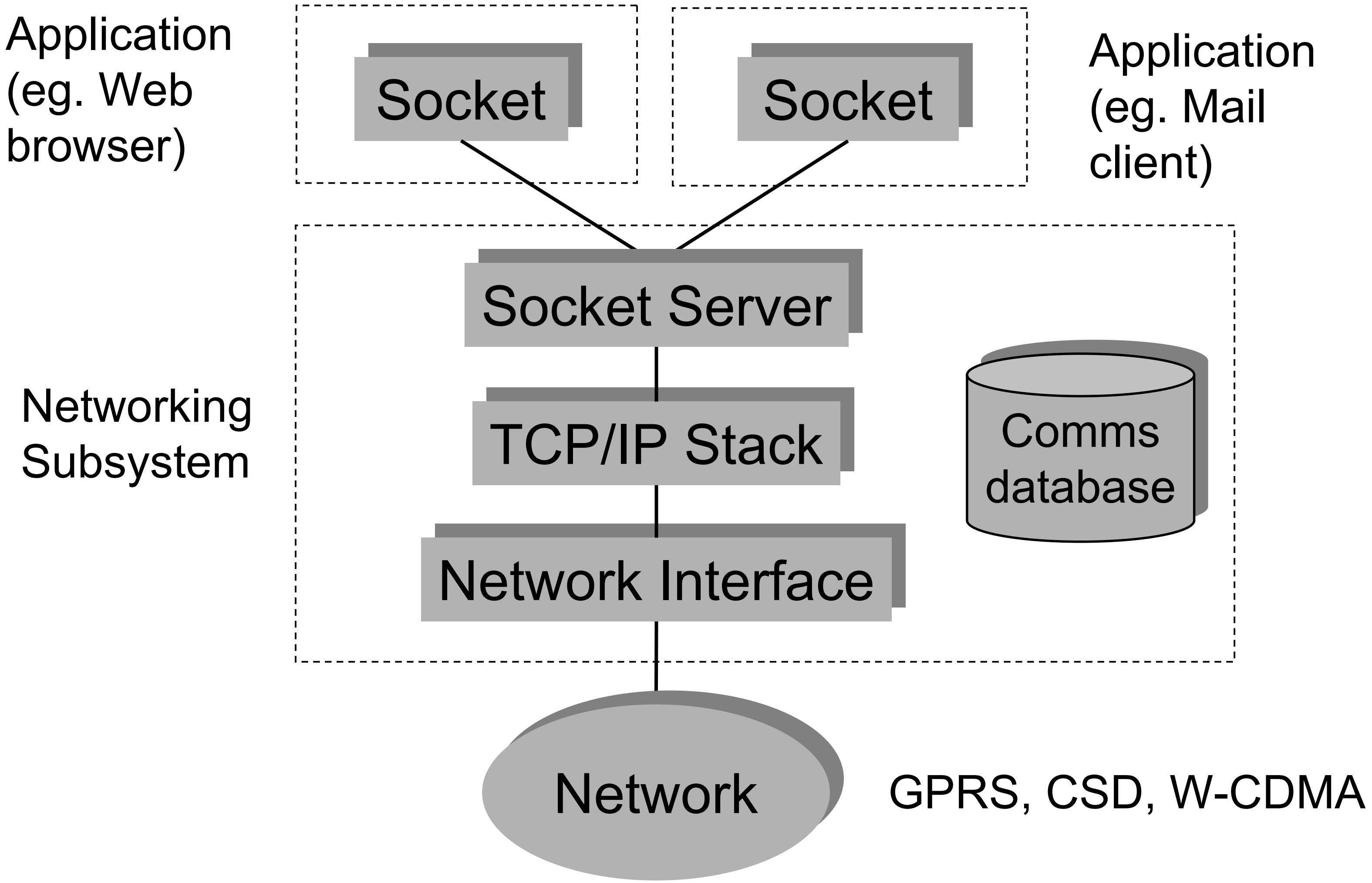
- applications access networking services through Berkeley sockets interface
- plug-in modules provide protocol implementations
  - …eg. hybrid IPv4/IPv6 TCP/IP stack
- network interfaces used to support various access technologies
  - …Circuit Switched Data (CSD - *dialup*)
  - …GPRS
  - …3G (W-CDMA, CDMA2000)
  - …Ethernet/IEEE802.3

Networking architecture

symbian exposium 03
definitive open environment

- communications database (commDB) used to store networking-related settings, including
  - …ISP phone numbers
  - …usernames and passwords
  - …GPRS settings

Networking Architecture



Background:
What is multihoming ?

Background | Implementation concepts | Application APIs

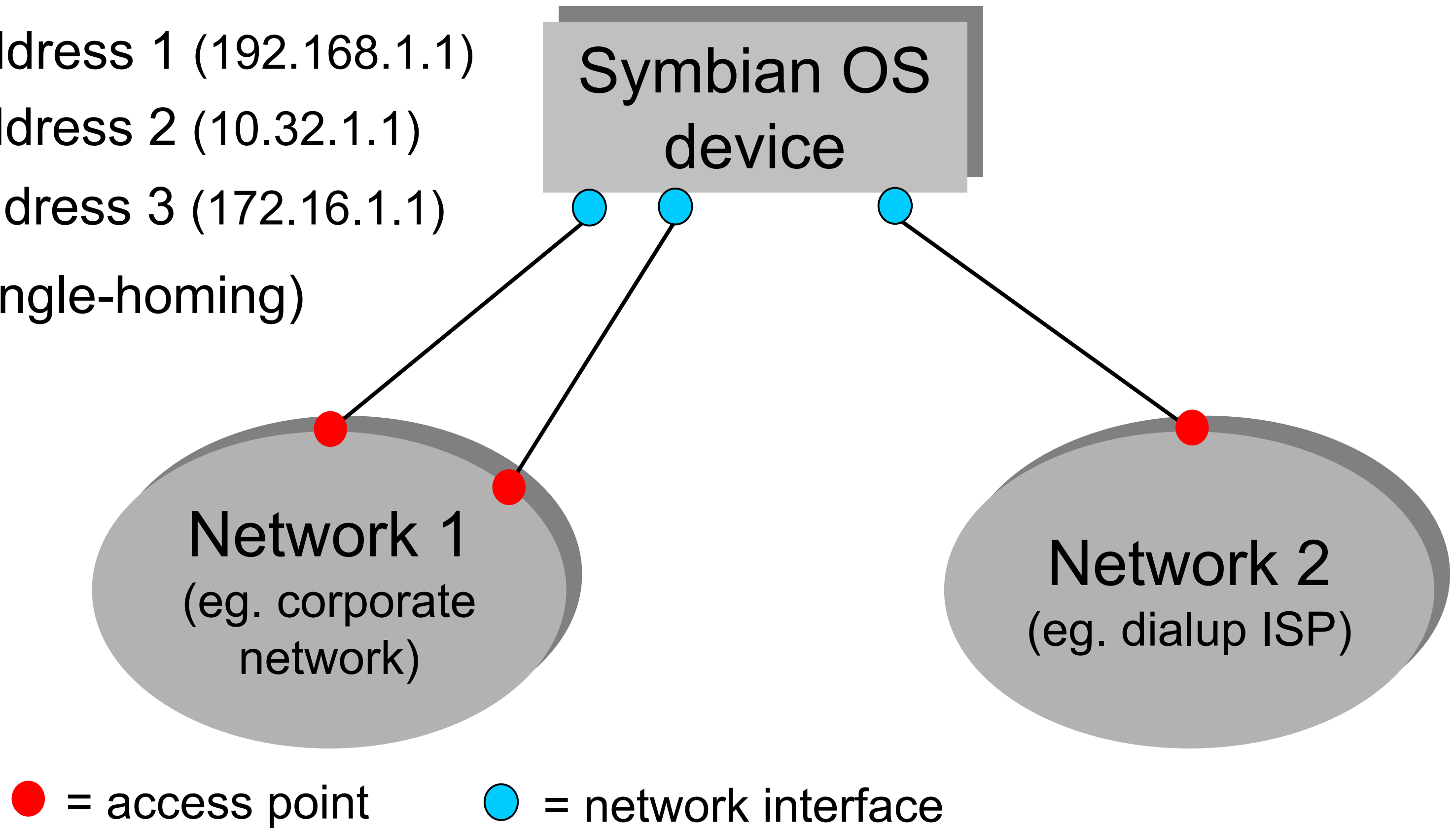
Multihoming definition

- simple definition
  - …a "multihomed" mobile device can support several network interfaces simultaneously
- previous versions of Symbian OS (v6.x and v7.0) were "single-homed"
  - …only supported a single network interface

Multihoming concept

IP address 1 (192.168.1.1)
IP address 2 (10.32.1.1)
IP address 3 (172.16.1.1)

(Single-homing)



Background:
Why is multihoming needed?

Why is multihoming needed?

- allows connection to multiple GPRS and W-CDMA (3G) services simultaneously
  - …eg. MMS, Internet, WAP
- allows future devices to connect via several different access technologies simultaneously
  - …eg. WLAN, CSD, W-CDMA (3G), Bluetooth

Why is multihoming needed ?
GPRS and W-CDMA Networks

- ...provides access to services via “Primary PDP Contexts”
  - ...represents a connection between a mobile device and the service on the network
  - ...has an *Access Point Name* to identify the service (eg. “internet”)
  - ...requires its own IP address

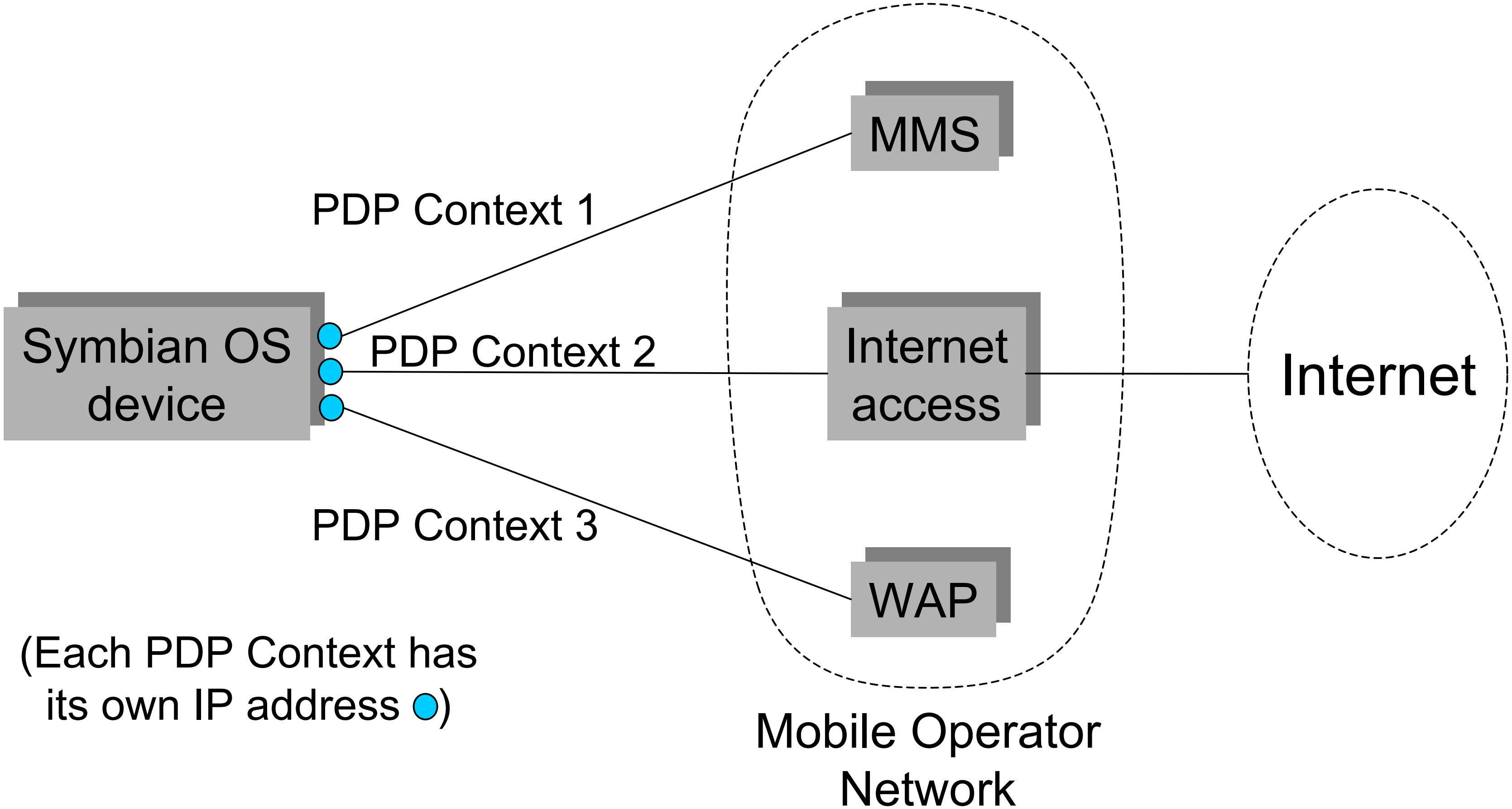
Why is multihoming needed ?
GPRS and W-CDMA Networks

- different services use different PDP Contexts

  …eg. MMS, Internet and WAP access

…so multihoming is required for
simultaneous access to multiple services

Accessing multiple services in GPRS and 3G networks



(Each PDP Context has its own IP address ●)

Implementation concepts: the challenges

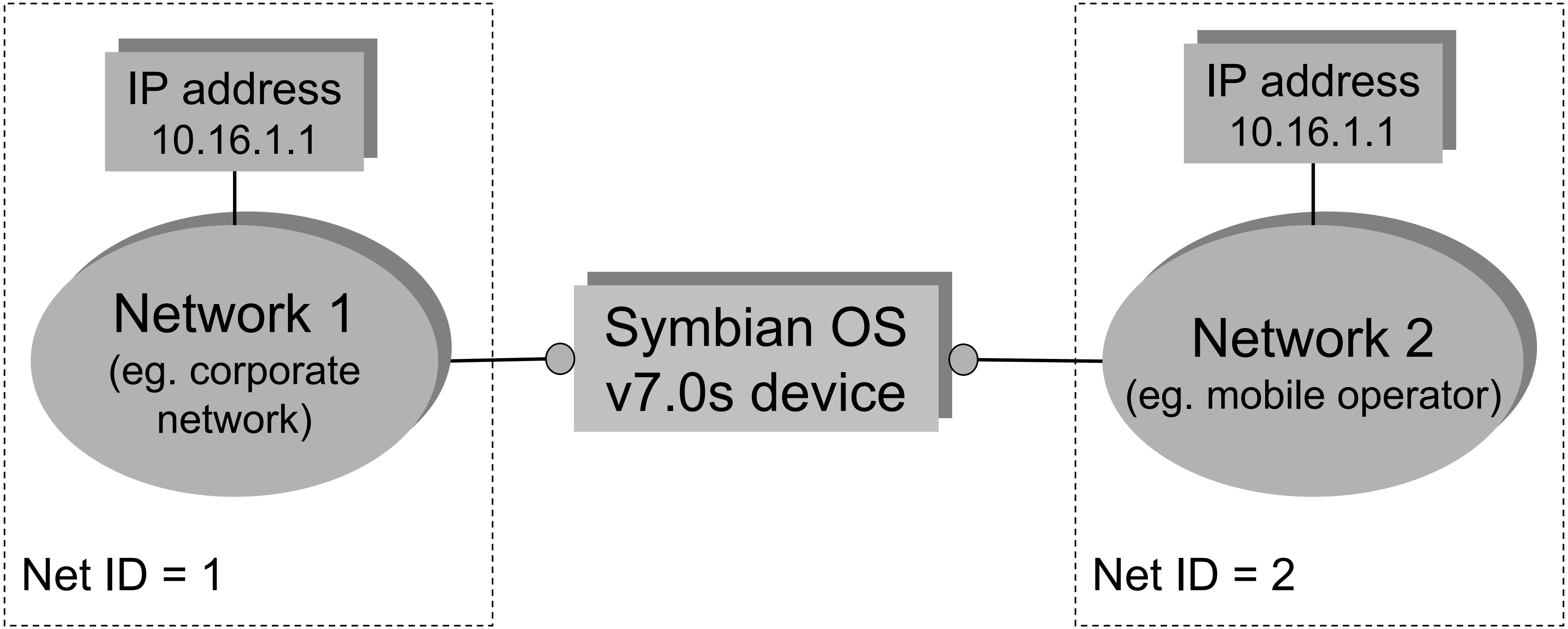
Multihoming Implementation Challenges

- managing duplicate “private” IP address ranges
  - …Two different networks can use the same private IP address range
- providing a suitable API towards applications
  - …allowing applications to:
    - manage multiple connections
    - associate sockets with particular connections

Multihoming Challenges:
duplicate IP addresses

- certain IP addresses are designated as “private” (RFC1918)
  - …10.x.x.x
  - …172.16.0.0 to 172.32.255.255
  - …192.168.x.x
- private IP addresses are not globally unique
  - …can appear in two different networks simultaneously
  - …can lead to ambiguity...

Multihoming Challenges:
duplicate IP addresses



How does the device route a packet destined for 10.16.1.1 ? Introduce network IDs

Multihoming Challenges: network IDs

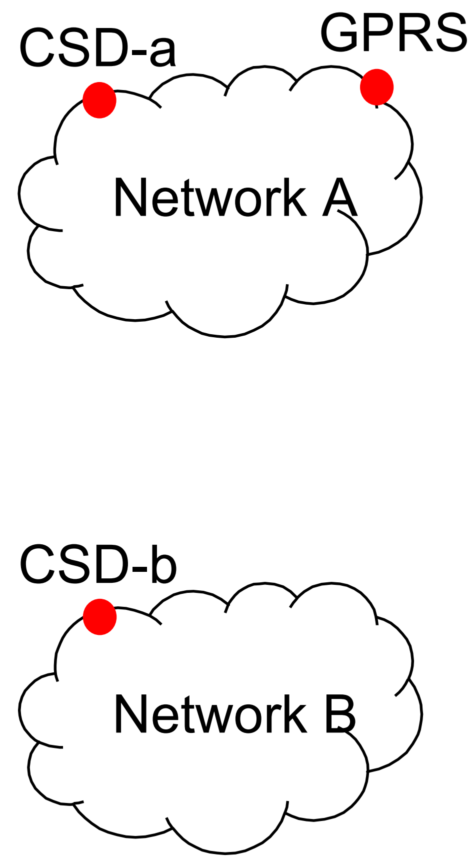
- applications can specify network ID (per socket)
  - …packets routed to specified network
  - …TCP/IP stack routes packets according to network ID as well as IP address
- network IDs also allow representation of multiple points of attachment to the same network
  - …eg. WLAN and CSD connections towards the same corporate network
- network IDs stored in comms database (CommDB)

Multihoming Challenges:
network IDs in CommDB

<u>CommDB (before v7.0s)</u>

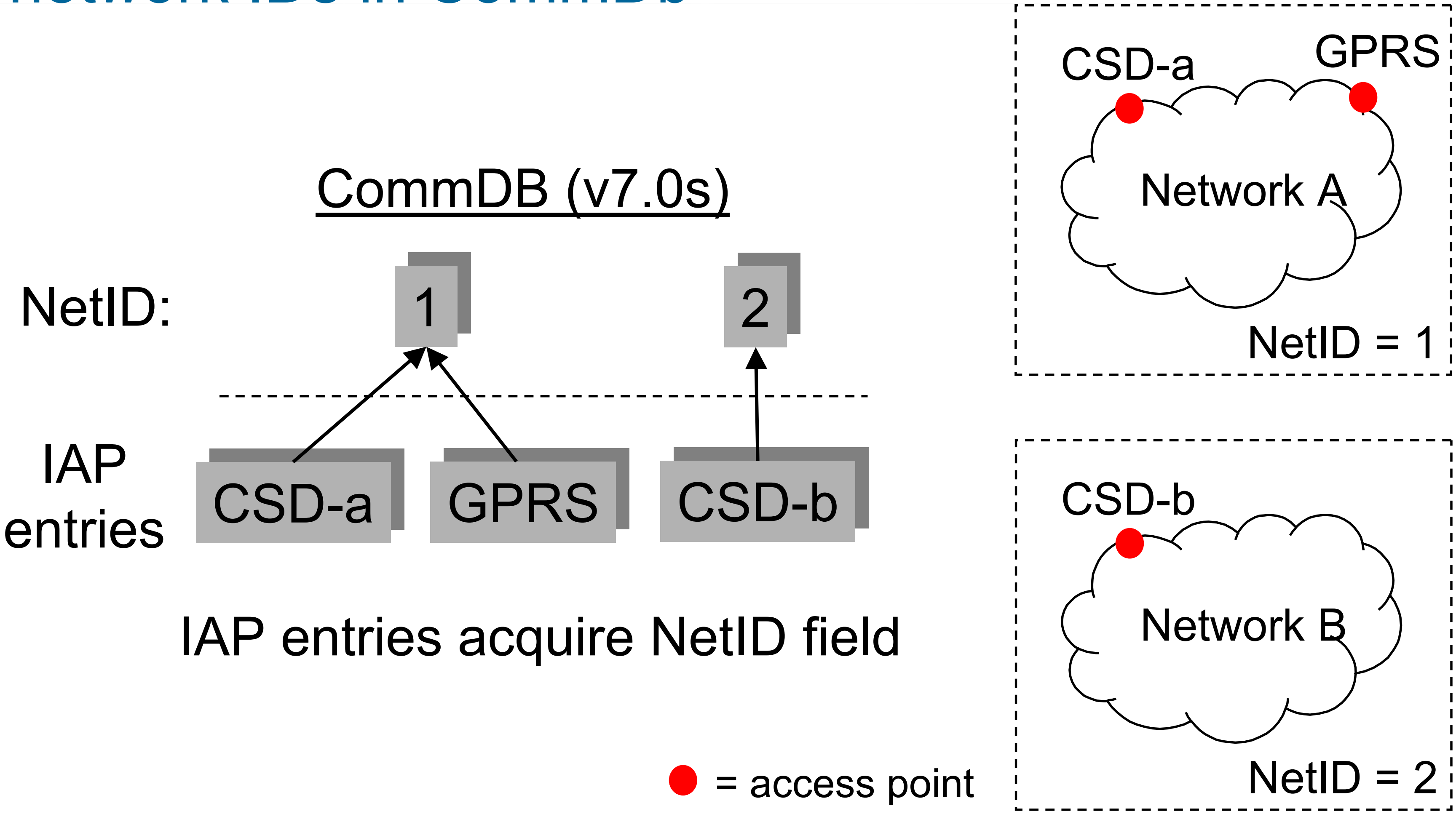
IAP entries

CSD-a | GPRS | CSD-b



= access point

Multihoming Challenges:
network IDs in CommDb



Application APIs: managing multiple connections

Background | Implementation concepts | Application APIs

Networking APIs before v7.0s (recap)

- networking APIs used by applications
  - ...RSocket
    - BSD sockets for data transfer
  - ...RHostResolver
    - name to address resolution (eg. www.symbian.com)
  - ...RGenericAgent
    - starting/stopping network interfaces

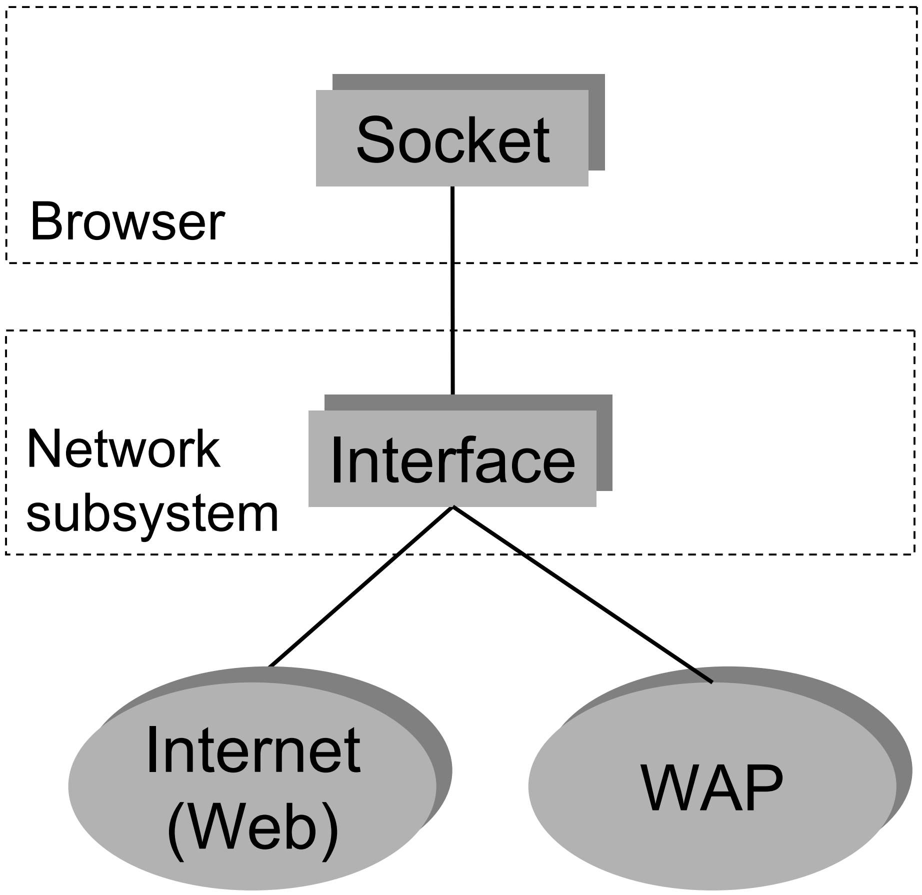
Networking APIs in v7.0s

- old APIs not suitable for multihoming
  - …RGenericAgent
    - designed for single interface model
  - …RSocket / RHostResolver
    - no control over destination network

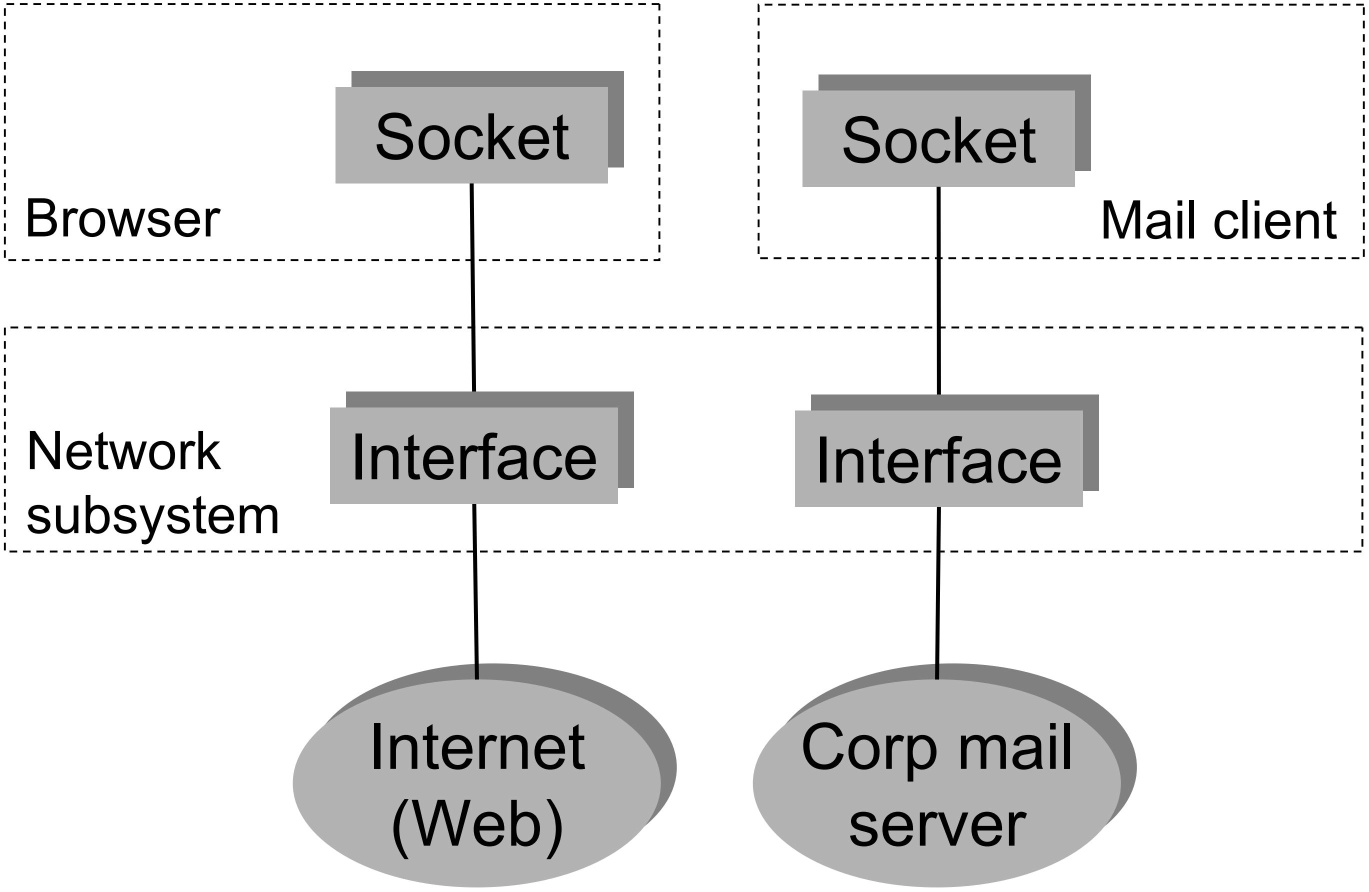
Networking APIs in v7.0s

- introduced new API - RConnection
  - …allows multiple connections to be managed
  - …sockets and host resolvers can be associated with a particular Network
  - …RGenericAgent deprecated
    - functionality incorporated into RConnection

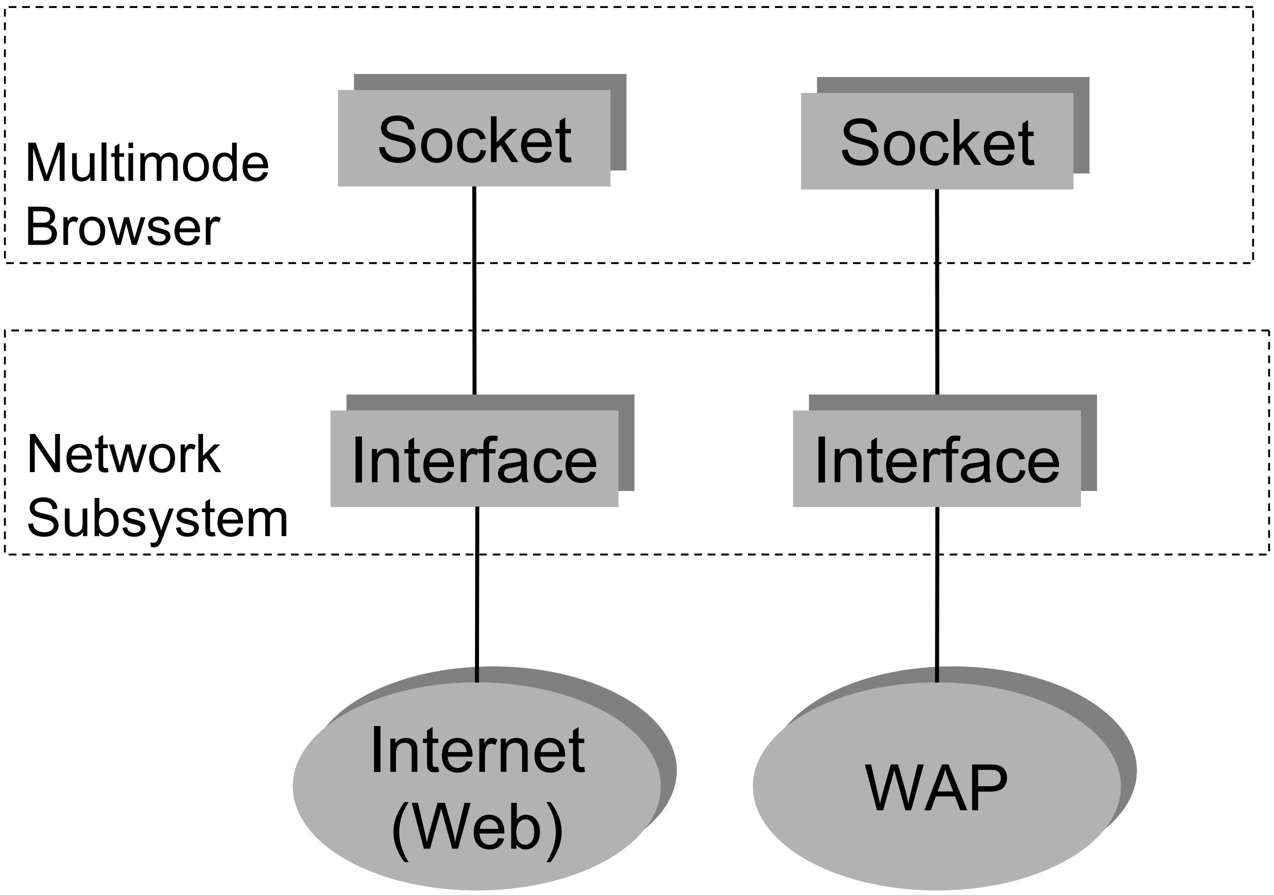
Singlehoming limitations: within an application



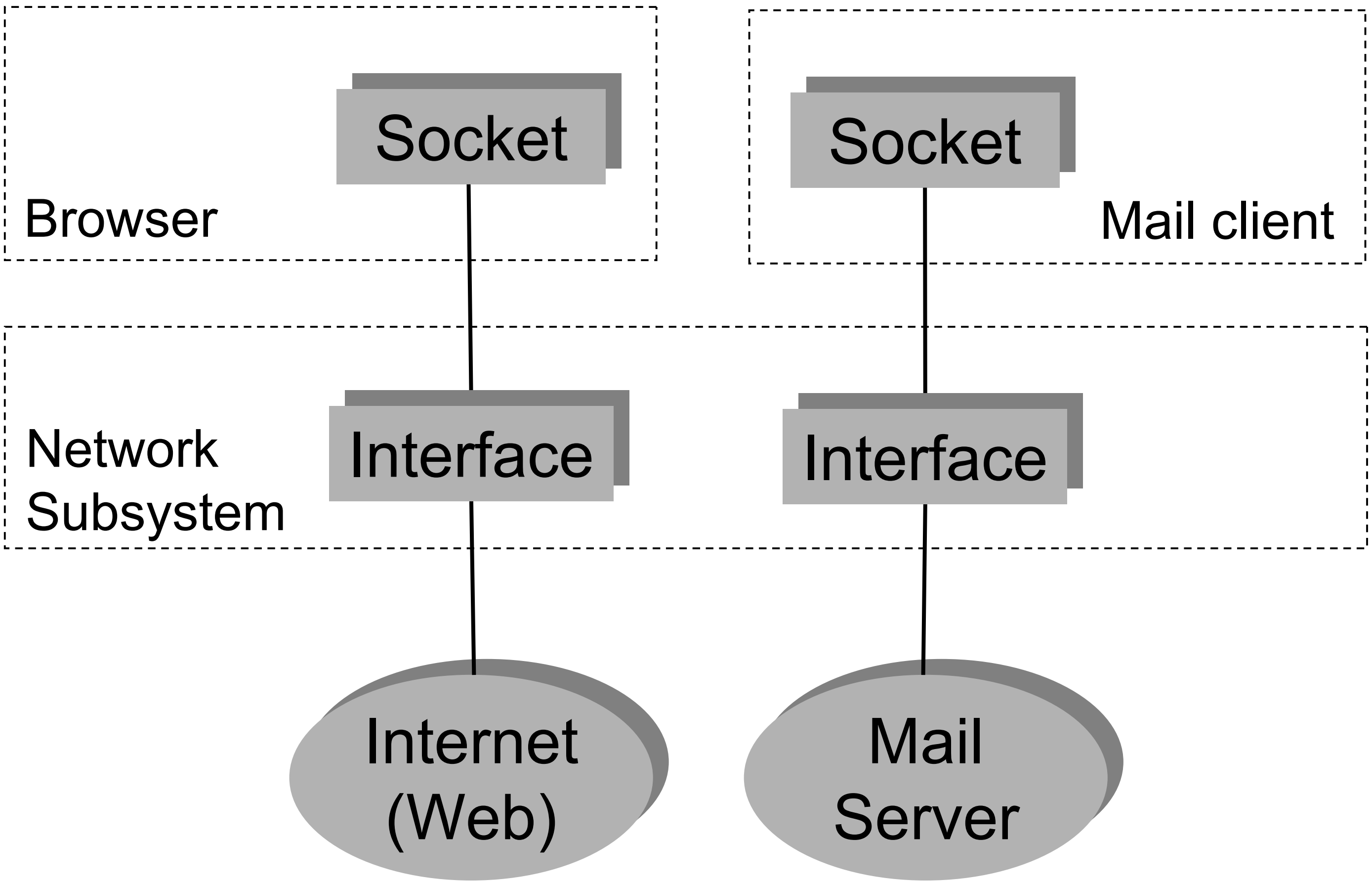
Singlehoming limitations: within the whole system



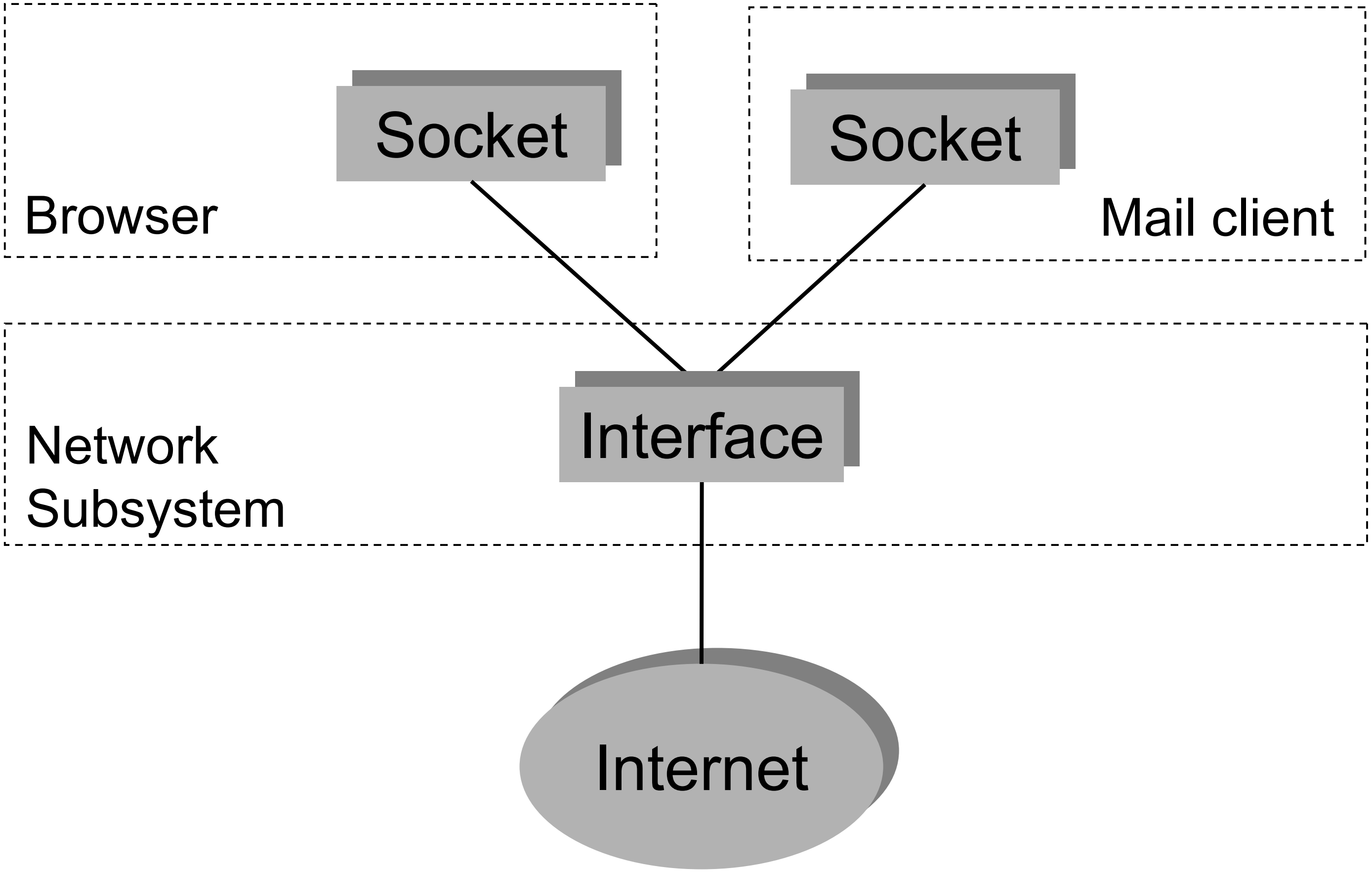
Multihoming possibilities: within an application



Multihoming possibilities: within the whole system



Multihoming: common use case



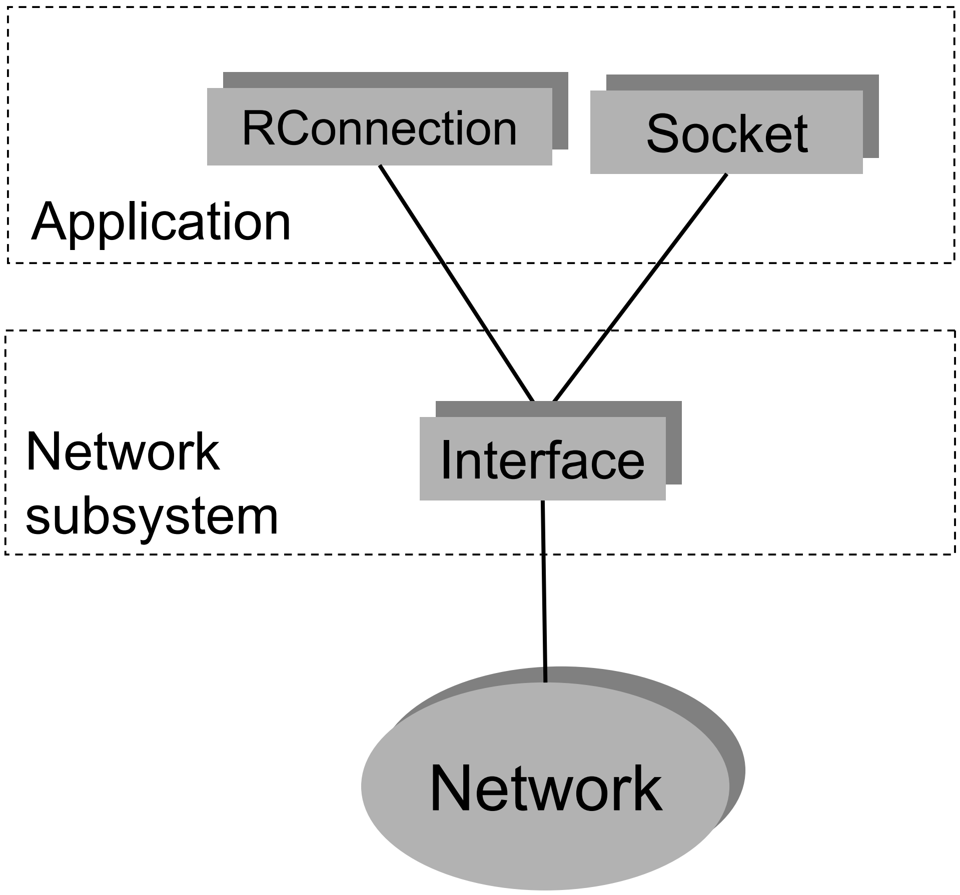
Networking APIs in v7.0s:
RConnection API details

- RConnection used to manage connections
  - …start / stop
  - …obtain progress notification as state changes (eg. link layer up/down)
  - …associate sockets and host resolvers with a connection
  - …access Comms Database (CommDB) settings for a connection

Networking APIs in v7.0s:
RConnection API details

- RConnection used to manage connections (cont)
  - ...indicate whether data is being transferred
  - ...enumerate active connections
  - ...obtain statistics
    - bytes sent / received
    - time connection was started

Example: Application using RConnection



Open RConnection

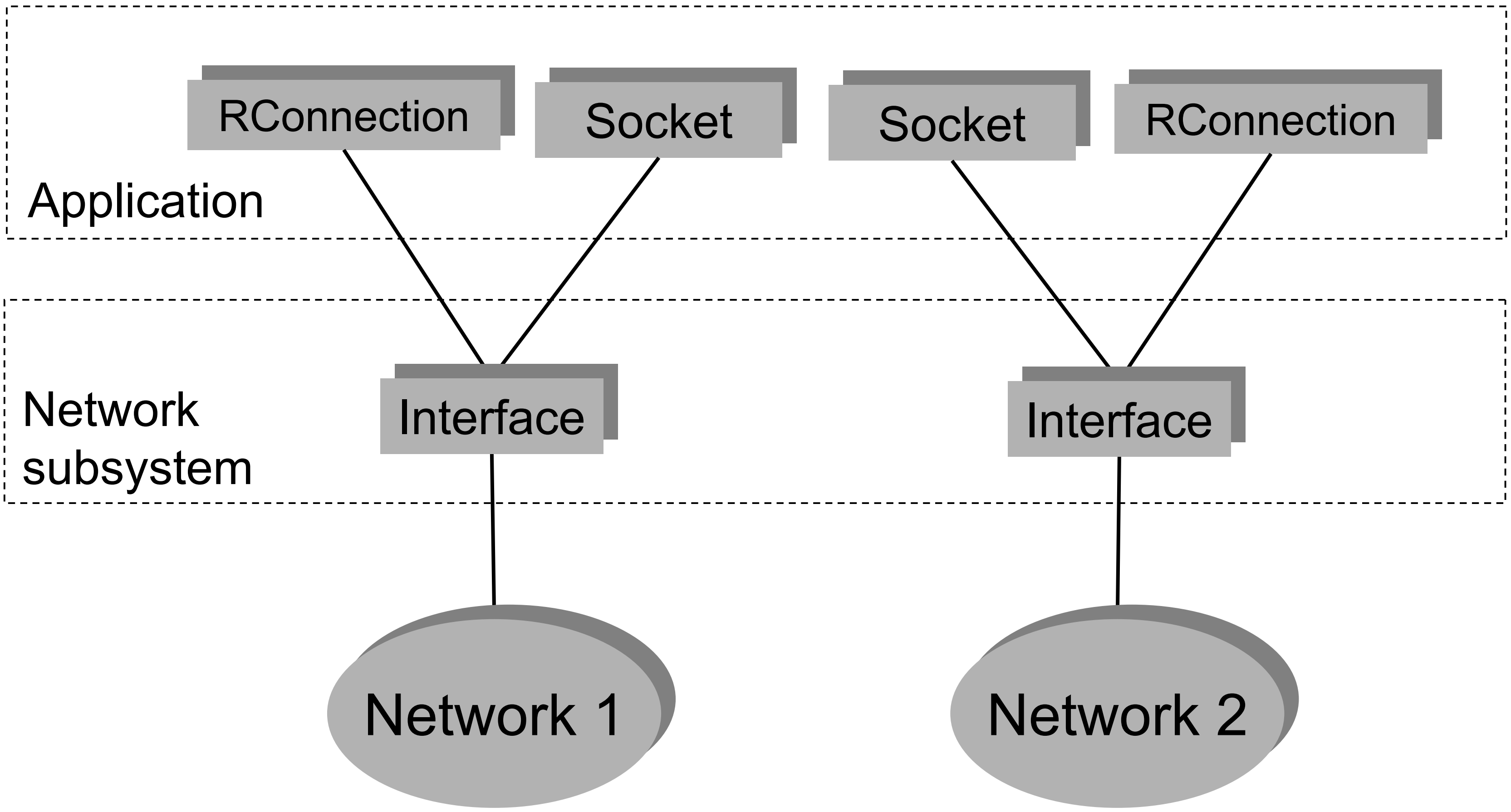
Start connection

Open socket

Associate socket with connection

Start data transfer

Example: Application using RConnection



Networking APIs in v7.0s:
Assumptions that are no longer true

- you can no longer get “*the* IP[v4] address” for the local device
  - …there might be several device IP addresses, depending on which interfaces are up

Networking APIs in v7.0s:
implicit connection

- in previous releases, it was possible to open a socket, or a host resolver, and "just use it"
  - …connection to the network was started automatically
- still possible in Symbian OS v7.0s, using the *implicit connection*
  - …preserves the old single homing behaviour for applications that need it
  - …old RGenericAgent API acts on implicit connection

Networking APIs in v7.0s:
implicit connection caveats

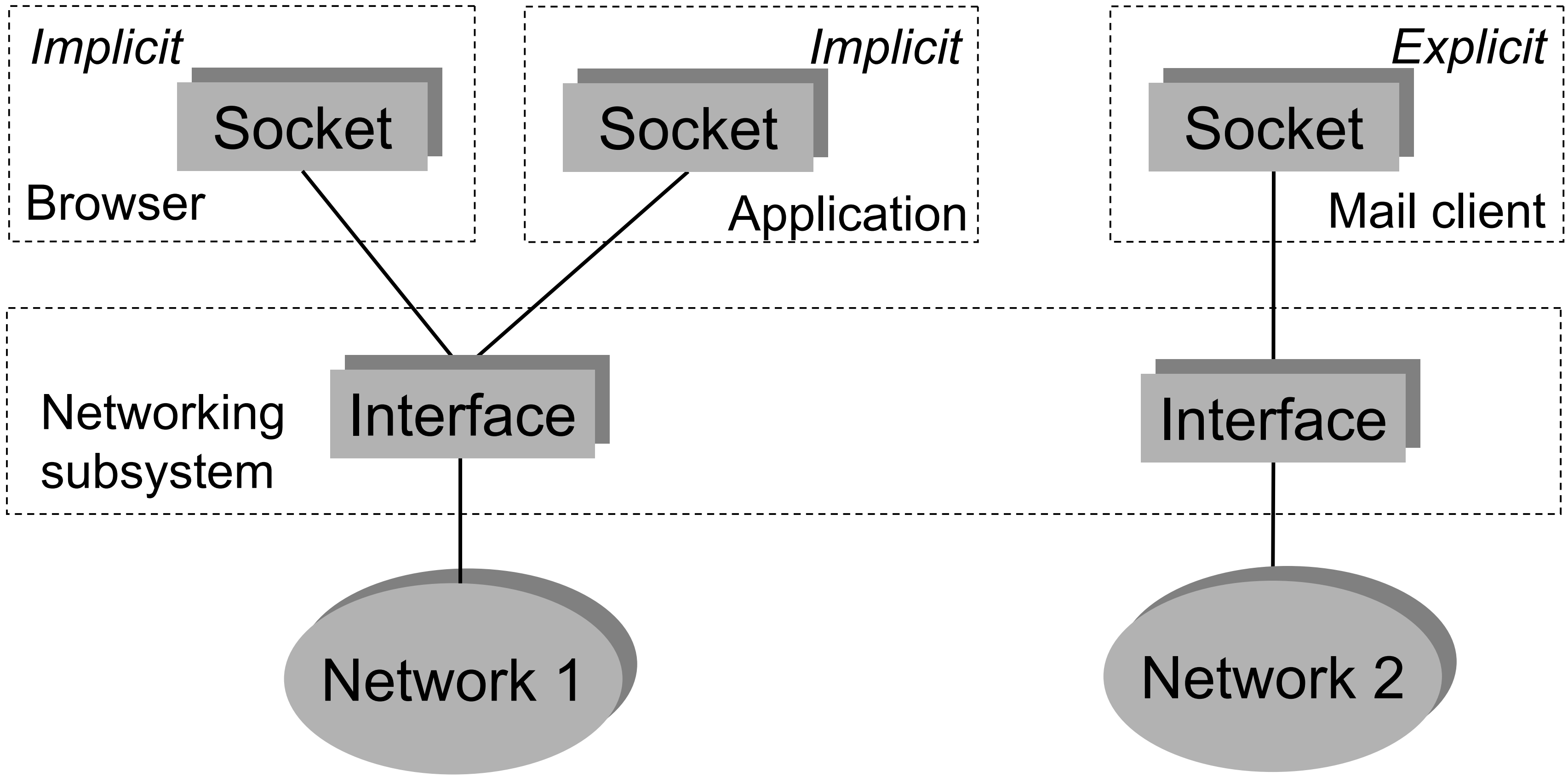
- application has no control over the connection to start
  - …depends on "Connection Preferences" in CommDB:
  - …either a designated connection
  - …or a connection selected by the user via a dialog box
- only one implicit connection on the system
  - …all applications share the same implicit connection
  - …if another application has caused it to be opened, you will get "their" connection

Networking APIs in v7.0s:
explicit connections

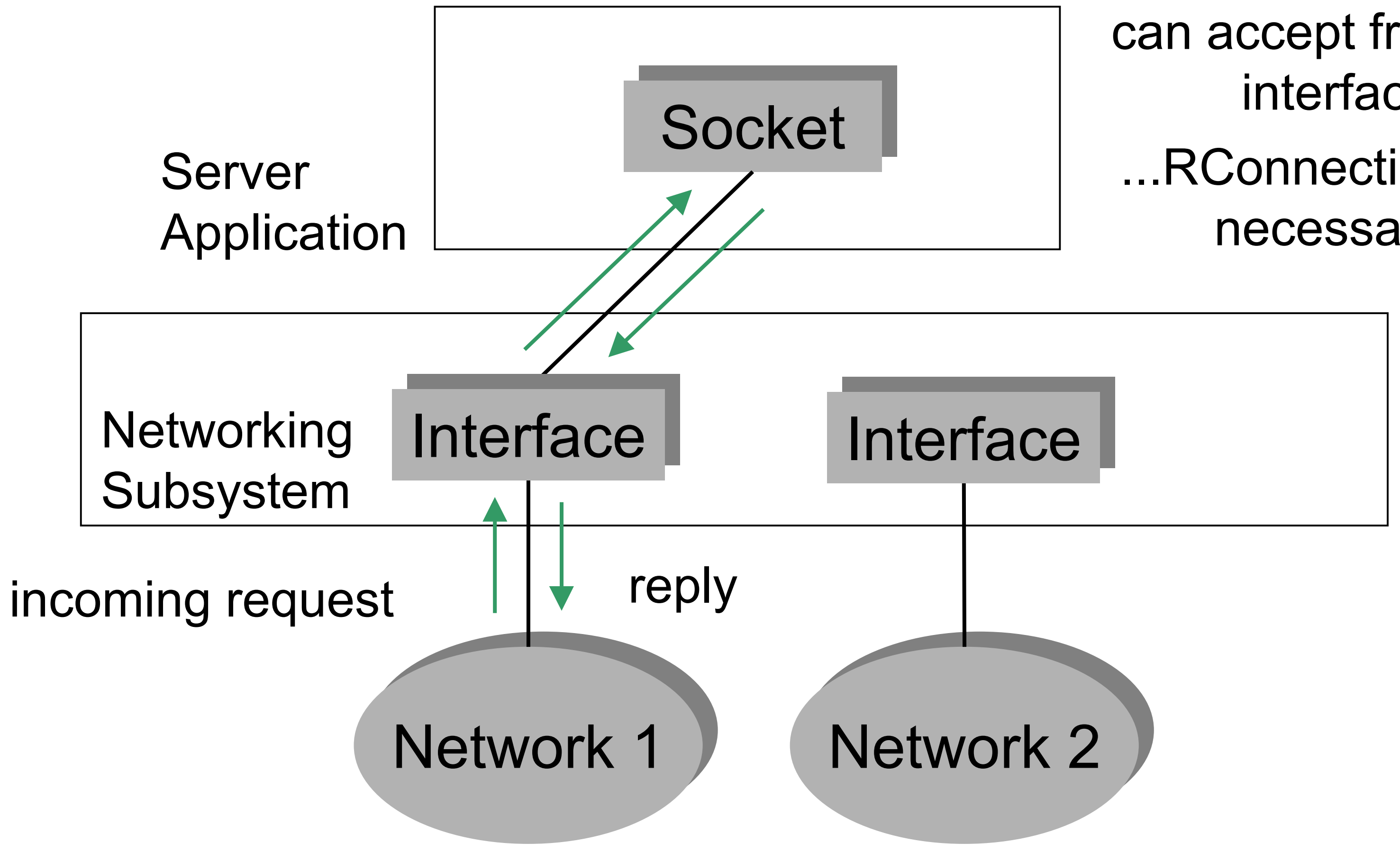
- *Explicit connections* are those that are started directly by the application, using the RConnection API

  …allow the application to explicitly specify the connection to start

Implicit connection example



Advanced topic:
servers and listening sockets



can accept from any interface

...RConnection not necessary

Symbian applications modified for multihoming

- HTTP framework
  - …additional property for specifying RConnection instance
- Messaging client
  - …modified to use RConnection
  - …future enabler for multiple interfaces

Summary

Summary

- previous versions of Symbian OS (eg. v6.1, v7.0) were single-homed
- need for multihoming driven by:
  - …ability in 3G Networks to access multiple services simultaneously (multiple PDP Contexts)
  - …need to enable future Symbian OS devices to connect to different media simultaneously if required (eg: WLAN, CSD, GPRS, WCDMA)

Summary (cont)

- introduced RConnection API
  - …allows control over multiple connections
  - …allows binding of sockets to networks
- “network ID” concept introduced
  - …removes ambiguity of duplicate IP addresses
  - …allows modelling of multiple interfaces to same network

Summary (cont)

- implicit and explicit connections
  - …single implicit connection - models old single homing behaviour (including RGenericAgent API)
  - …multiple explicit connections - started via RConnection

Questions